

AI & Technology Landscape — August 2026

Frontier models, CAD tooling, PDM build notes and estimating-adjacent software — mapped to what we are actually building

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This month the report widens. July covered frontier models only; this one adds **the CAD and design tooling Yogesh is working alongside, build notes for the in-house PDM, the SOLIDWORKS partner marketplace, and the tooling layer inside the estimating engine itself** — because the useful question is no longer "which model is best" but "what is now buyable, buildable, or already sitting inside a licence we hold".

Two entries below are worth your time before the rest. **SOLIDWORKS LEO already does PDF-to-parametric-3D**, which is F1 on the technical design roadmap and currently marked "POC pending" — it may not need building. And **Qwen3.8-27B is an open-weight vision model that fits a single 24 GB card**, which changes what on-premise drawing vision costs from pence-per-call to nothing.

On sourcing. Entries marked **VENDOR** come from the vendor's own documentation. Entries marked **SECONDARY** come from third-party write-ups and should be confirmed against vendor pricing pages before anything is committed — figures in this class move, and several of the August pricing numbers are introductory rates with end dates.

THE MONTH IN ONE SCREEN What shipped, and whether it changes anything for us

DATE	WHAT	WHO	WHY IT MATTERS HERE	VERDICT
12 Aug	Grok 4.6	xAI	Multimodal across text, image and audio; 500K context; built for long-running agents. Direct successor to the Grok line the engine already calls for drawing vision.	ONE-LINE SWAP
13 Aug	Gemini 3.7 Flash	Google	Introductory \$0.75/\$3.75 per 1M to 31 Dec 2026, then	TRIAL

DATE	WHAT	WHO	WHY IT MATTERS HERE	VERDICT
			\$1.50/\$7.50. Cheapest capable vision route for rasterised drawing pages.	
14 Aug	Qwen3.8-27B	Alibaba / Qwen	Apache-2.0, 27B dense, vision encoder , 262K native context. Runs 4-bit on a single 24 GB card. No per-call cost, no drawings leaving the building.	LOOK AT THIS
24 Jul	Claude Opus 5	Anthropic	\$5/\$25 per 1M, 1M context, 128K output, effort toggle. Near-Fable-5 capability at half Fable's price. Already wired — <code>_call_anthropic_llm</code> exists.	SWAP NOW
2026 SP1	SOLIDWORKS LEO	Dassault	Agentic assistant in the Labs tab. Converts a 2D drawing or PDF into an editable parametric model; auto-generates drawings; rebuilds STEP/IGES as parametric parts.	CHECK OUR SEAT
—	GPT-5.7	OpenAI	Does not exist. GPT-5.6 (9 Jul) remains current; Luna became the ChatGPT free-tier default during August. Nothing to act on.	NO CHANGE

THE ONE THAT CHANGES THE SHAPE OF THE PROBLEM — ON-PREMISE VISION

QWEN3.8-27B · APACHE-2.0 · RELEASED 14 AUG 2026

Why this is first and not buried under the frontier models

Every drawing-vision call the engine makes today leaves the building and costs money per page. That is a per-job cost, a per-job latency, and a standing question about sending customer drawings to a third party. Qwen3.8-27B is the first open-weight model that plausibly removes all three at once: **Apache-2.0 licence, a built-in vision encoder, and small enough to run on one graphics card.**

SDI already runs its own tin. This is the kind of thing that fits how the business is set up, not a subscription bolted onto it.

What it is

SECONDARY

27-28B dense

Vision encoder

262K native context

Apache-2.0

Announced 3 August alongside Qwen3.8-Max, weights published 13-14 August on Hugging Face under Apache 2.0. A dense 27B model with hybrid attention and — the part that matters for us — a **vision encoder**, so it reads images, not just text. Native 262K context, reported extensible to 1M.

Reported benchmarks put it at 61.7 on SWE-bench Pro and 52 on the Artificial Analysis Intelligence Index, a 14-point jump over Qwen3.6-27B. Simon Willison's write-up is positive but flags that it *defaults to wildly overthinking things* — worth knowing before you measure latency.

What it would take to run here

Full BF16 weights are around 55.6 GB. **Quantised to 4-bit it fits a 24 GB card — an RTX 3090 or 4090.** Reported throughput on modest hardware: 24.5 tokens/sec on a Ryzen AI Max+ 395, 51.8 tokens/sec on a Radeon AI PRO R9700 with 32 GB.

That is one workstation-class card, not a server rack. Against the current estimating box — which already needs a machine capable of running SOLIDWORKS COM — this is an incremental purchase rather than a new class of infrastructure.

Where it would slot into the engine

Source rank 40

Drawing vision

Note scan

The price chain currently puts **vision at rank 40** — the weakest source, used to fill gaps the deterministic readers leave. That is exactly the right place to trial a local model: the arbitration already treats its output as low-confidence and will not let it override a DXF or a title-block read.

The honest test: take the 10575-02 pack, run the same pages through the current vision path and through a local Qwen3.8, and compare against the manual estimate. If it holds, the per-job vision cost goes to zero and the drawings stop leaving the network. If it does not, we have lost a day and learned the ceiling.

→ **NEXT: ONE CARD, ONE AFTERNOON, 10575-02 AS THE BENCHMARK — BEFORE COMMITTING TO ANYTHING**

The caveats, stated plainly

Everything above is from secondary sources. Benchmark figures, quantised memory footprints and throughput numbers all come from third-party write-ups, not from a vendor page. Confirm against the official model card before buying hardware.

A 27B model is not Opus 5. It will be worse at ambiguous drawings, and the failure mode of a local model is the same as a hosted one — a confident wrong answer. Rank 40 and the credibility gate are what make trialling it safe.

Running your own inference is a job. Model updates, driver updates, and somebody who owns it. That is a real cost against "no per-call fee", and on a team where one developer is already the constraint it deserves saying out loud.

FRONTIER MODELS — WHAT ACTUALLY SHIPPED

Mid-August was the densest release window of the year

Claude Opus 5

24 JUL 2026

\$5 / \$25 per 1M

1M context

128K output

Effort toggle

Anthropic's near-frontier model: roughly Fable-5-level capability at half Fable's price, same sticker as Opus 4.8. API string `claude-opus-5`. 1M context both default and maximum; 128K output in standard requests, up to 300K through the Batches API with a beta header.

The new thing is a **low / medium / high effort toggle per request** — you trade cost against capability call by call rather than picking a model tier up front. For the engine that maps cleanly: low effort for high-volume classification, high effort for the handful of parts that decide whether an estimate is right.

Why this is the cheapest change on the page: `_call_anthropic_llm` already exists in `web_ai_price_lookup.py` and the key is already configured. This is a model-string change, not an integration.

→ **NOW: POINT THE ANTHROPIC PATH AT CLAUDE-OPUS-5; RE-RUN THE DETERMINISM PROOF**

Grok 4.6

12 AUG 2026

Multimodal: text · image · audio

500K context

Long-running agents

Direct successor on the line the engine already uses for drawing vision. Multimodal across text, image and audio, 500K context, positioned for long-running agent work and reported around half the price of rival frontier models.

Same API shape and auth as the Grok model already configured, so this is the other one-line change available this month. **Do the two swaps separately** — changing the vision model and the Anthropic model in the same run means a shifted number tells you nothing about which caused it.

Unverified: UK/EU API availability was an open question on the previous Grok release and has not been re-confirmed here. Check `console.x.ai` before deploying.

→ **NOW: BUMP THE XAI MODEL STRING; RE-RUN 10575-02 THREE TIMES; CONFIRM ITEM COUNTS HOLD**

Gemini 3.7 Flash

13 AUG 2026

\$0.75 / \$3.75 intro to 31 Dec 2026 then \$1.50 / \$7.50

Google cut 3.6 Flash to the same introductory rate when 3.7 launched, so both sit at \$0.75/\$3.75 per 1M until the end of the year and revert to \$1.50/\$7.50 after. 3.6 Flash (21 Jul) remains a listed stable model — 1,048,576 input tokens, 65,536 output — and is the sensible target if you want a version that will not move under you while testing.

The reason to care is unchanged from July: **native image input**. Rasterise a drawing page at 300 DPI, send it, ask for structured JSON. The model sees the drawn flat-pattern, the dimension callouts and the fold arrows rather than inferring from vector paths.

Note the date. Introductory pricing to 31 December means any cost case built on \$0.75 needs a line saying what happens in January.

→ **COMPARE HEAD-TO-HEAD WITH LOCAL QWEN3.8 ON THE SAME PAGES BEFORE CHOOSING A VISION ROUTE**

OpenAI — nothing new

NO ACTION

There is no GPT-5.7. GPT-5.6 from 9 July remains current across Sol, Terra and Luna. The only August movement was Luna becoming the ChatGPT free-tier default, which is a consumer change and does not touch us.

Forecasters put Q3 2026 as the heaviest frontier release window of the year, with several launches expected between mid-August and late September. Three have now landed. If OpenAI is in that group, the next report is where it shows up.

CAD AND DESIGN TOOLING — INSIDE THE LICENCE WE ALREADY HOLD

SOLIDWORKS 2026 shipped three AI companions. One of them overlaps the roadmap directly.

F1 may already exist

The technical design roadmap carries **F1 · PDF → ASM/PRT reconstruction** as *POC pending*, with dates to be confirmed — rebuilding 3D parts and assemblies from PDF drawings, described as the furthest-reaching item on the list.

LEO does this today. It converts a 2D drawing or PDF into an editable parametric model, and it is shipping in SOLIDWORKS 2026 SP1 through the SOLIDWORKS Labs tab in the Task Pane. Before any work is scoped against F1, somebody should open that tab on a seat we already pay for and try it on a real legacy drawing. If it works even partially, F1 changes from a build to an evaluation.

LEO — the engineer's companion

VENDOR

SW2026 SP1 / FD01

Labs tab

Agentic

Reverse-engineers BREP bodies and 2D drawings into parametric models. Generates manufacturable parts from sketch inputs. Builds assemblies from natural-language prompts. Auto-generates drawings — with control over sheet sizes, drawing standards and which template to use. Sets up SIMULIA Abaqus simulations from a prompt. Will attempt to rebuild a single-body STEP or IGES as an editable parametric part.

Overlaps two roadmap items, not one. The PDF-to-parametric work is F1. The drawing generation overlaps **C3 · Drawing generation from ASM/PRT**, which is already POC-complete in-house and waiting on a licence. Worth understanding which of the two SDI would rather own, because they are not both needed.

→ **THIS WEEK: OPEN SOLIDWORKS LABS ON AN EXISTING SEAT; TRY ONE LEGACY PDF; REPORT BACK**

AURA — and what it deliberately cannot do

VENDOR

AURA is the knowledge and governance companion — connects SOLIDWORKS users to company knowledge via the 3DEXPERIENCE platform, and can explore and analyse web knowledge.

The important line for us: AURA *does not access your PDM or PLM vault data* and does not retrieve engineering knowledge from your organisation's history. That is a direct answer to a question the PDM phases would otherwise raise — Dassault's own AI is not going to read the vault for you, so P3.1–P3.4 are not about to be made redundant by the tool vendor.

MARIE — probably not for us

VENDOR

The scientific companion: materials science, hypothesis development, microbiology, chemistry, laboratory analysis. Aimed at design validation with deeper scientific reasoning.

Recorded for completeness. Retail and hospitality display work does not obviously need a microbiology companion, and nothing here suggests chasing it.

BUILD NOTES FOR THE IN-HOUSE PDM

P3.1 starts 8 September. One known problem is worth designing around now, not in October.

SolidWorks files churn at the binary level — every open and save

The single most repeated warning in the CAD-version-control literature is this: **a SOLIDWORKS assembly changes at the binary level every time it is opened and saved, even if nothing was moved.**

For a PDM built on GitLab and LFS that has a direct consequence. Every open-and-close produces a new LFS object of full file size, with no meaningful diff. Storage grows with *activity* rather than with *change*, and history fills with revisions that record nothing. On a vault of assemblies that compounds quickly.

This is a design question for P3.1, not a defect. It is answerable — content-hash the meaningful payload rather than the file, or gate commits on an explicit check-in rather than on save, or prune unreferenced LFS objects on a schedule. But it is far cheaper to answer in the MVP than to retrofit once there is a year of vault history.

Prior art worth an hour

SECONDARY

Not as alternatives to building — the in-house decision is made — but because each has met the same problems and their documentation says how they landed:

- **Anchorpoint** — Git plus LFS positioned explicitly as a lightweight SOLIDWORKS PDM alternative for smaller teams. Closest thing to the architecture P3.1 describes.
- **DocDokuPLM** and **openPLM** — open-source PLM. Heavier than needed, but their data models are worth reading before designing ours.
- **lakeFS**, **git-annex**, **ArtiVC** — large-file versioning approaches other than Git LFS, if the churn problem above bites harder than expected.
- **SVN with TortoiseSVN** — the old answer. Reliable, no CAD intelligence, manual link management. Useful mainly as the baseline the new thing has to beat.

→ **BEFORE P3.1 DEVELOPMENT STARTS ON 16 SEP: READ HOW ANCHORPOINT HANDLES BINARY CHURN**

What this does not change

The case for building rather than buying is unchanged and is worth restating, because a landscape report is exactly where somebody reasonably asks "why are we writing our own?".

The in-house PDM is being built with a **Document Manager intelligence layer over PostgreSQL** — it knows what is inside the files, not just that they changed. Nothing on the list above does that, and it is the part that feeds the estimating engine. A generic version-control product would solve check-in and check-out and leave the reason we are doing this untouched.

ESTIMATING-ADJACENT SOFTWARE — WHAT THE MARKET HAS, AND WHAT IT DOES NOT

Worth knowing what exists before anyone asks why we built our own

The AI takeoff market

SECONDARY

Togal.AI Quotr.ai TheTakeoff.AI Autodesk Takeoff

A real and maturing category: computer vision reading drawing sets and pulling priced quantities automatically. Togal.AI classifies rooms and measures footprints from floor plans. Quotr.ai claims 95–99% vector accuracy feeding a live cost matrix. TheTakeoff.AI reads a full drawing set and pulls line lists, valve counts, instrument counts. Autodesk Takeoff does quantities from 2D sheets and 3D models inside Construction Cloud.

All of it is construction. Floor plans, rooms, valves, instruments — quantities from architectural and MEP drawing sets. None of it does sheet-metal flat patterns, bend counts, weld sequences, powder coverage by mass, or a five-tier price chain against a company's own historical job corpus.

What that tells us

Two things, and they point in opposite directions.

The approach is validated. Reading drawings with vision and producing priced quantities is not speculative — there is a competitive market doing it at scale in an adjacent industry. That is reassuring about the direction.

Nobody is coming to do ours. The gap between "measure a floor plan" and "cost a laser-cut, folded, welded and powder-coated display unit against 1,982 historical jobs" is the whole product. No vendor in that list is going to close it, which means the build-versus-buy question does not really exist for us — but it also means the accuracy work is ours alone.

Research worth tracking

SECONDARY

Academic work on **fine-tuned document-understanding transformers for parsing engineering drawings into structured information** is active and published. This is closer to our problem than the construction takeoff vendors are — parsing a title block, a BOM table and a revision block into fields is exactly what [extractor_patterns.py](#) does with regexes today.

Not an action. A direction: if the deterministic readers plateau, a small fine-tuned document model over SDI's own drawing corpus is the route that does not depend on any vendor.

THE WIDER SOLIDWORKS MARKETPLACE — WHAT IS BUYABLE RATHER THAN BUILDABLE

Partner add-ins and native tools that overlap the technical design roadmap

An add-in already batch-exports DXF with bend lines

The estimating engine's largest single source of inaccuracy is drawings arriving without flat patterns. **CADshift is a SOLIDWORKS add-in that batch-exports laser-ready DXF files with bend lines, metadata and BOM quantities** — which is the output C1 was built in-house to produce.

It does not dodge the licence problem. An add-in runs inside SOLIDWORKS and needs a seat, exactly as C1 does. So this is not a way around the constraint. What it might be is a way to stop *maintaining* that capability ourselves — the seat gets resolved once and then the flat-pattern export is somebody else's product to keep working.

TOOL	WHAT IT DOES	OVERLAPS	WORTH A LOOK?
CADshift	Batch DXF export with bend lines , metadata and BOM quantities. Laser-ready output from assemblies.	C1 · export-dxf	YES — PRICE IT
SOLIDWORKS Costing	Native. Model-based manufacturing cost estimates from the part geometry. Included, not an add-on purchase.	The estimating engine, partially	KNOW WHAT IT DOES
DriveWorksXpress	Native. Automates repetitive design tasks and product variants from rules.	Configurable display families	ALREADY OWN IT
BatchProcess	Export to 28+ formats — DXF, DWG, IGES, STEP — plus automatic printing and custom-property management.	C1 · export-package	COMPARATOR
ToolXperts	DXF generation, drawing creation, BOM	C1, C3 · and Phase 4 nesting	COMPARATOR

TOOL	WHAT IT DOES	OVERLAPS	WORTH A LOOK?
	calculations and linear nesting .		
CAMWorks	Fully integrated CNC programming inside SOLIDWORKS.	Workstream 4 · CNC / G-code	RELEVANT LATER
Xometry Add-In	Instant quotes and lead times without leaving CAD — for parts bought out rather than made.	Bought-in pricing	NOTE ONLY

SOLIDWORKS Costing — worth understanding before anyone else finds it

Dassault ship a costing tool in the box. Somebody will eventually ask why SDI built an estimating engine when SOLIDWORKS does costing, so it is worth having the answer ready.

SOLIDWORKS Costing prices **one modelled part from its geometry against a rate template**. It needs a model. It does not read a PDF pack, does not resolve a GA-to-sub-assembly-to-detail BOM tree, does not price against 1,982 historical SDI jobs, does not carry a five-tier source waterfall with confidence per line, and does not mirror the SDI estimating workbook to the cell.

It is a calculator; the engine is an estimator. Both true, and the distinction only sounds defensive if it has not been said in advance.

The honest read on buy-versus-build here

Nothing in this table replaces the estimating engine. Several things in it overlap **C1 and C3**, which are POC-complete in-house and licence-blocked.

That is the interesting question, and it is Yogesh's to answer rather than mine: given four finished tools waiting on a seat, is the flat-pattern export worth owning, or is it worth buying so the effort goes to the parts nobody sells — the PDM, the DXF generation from a GA where no model exists, and the estimating integration?

Nobody sells that last one. Which is roughly the argument for buying the first.

→ **WORTH 20 MINUTES WITH YOGESH BEFORE P2 DEVELOPMENT STARTS ON 27 AUG**

Paperless Parts — the closest thing to a competitor

SECONDARY

62 geometric attributes

Sheet metal specific

\$300-750 / month

Unlike the construction takeoff vendors, this one is **actually in our industry**. It detects 62 geometric attributes from a 3D CAD model, unfolds parts to get dimensions and thickness, and identifies cuts, bends, pierces, cut length and etchings as costing variables. It quotes assemblies including underlying components and purchased hardware as a single line rather than part by part. Growth tier \$300/month, Professional \$750/month.

Read that list again against our own faults. Bends, cut length, unfolding, assemblies-with-bought-ins — that is the same problem space, solved commercially.

And here is why it does not solve ours. It prices *from a 3D CAD model*. SDI's defining problem is the packs that arrive *without* models or flat patterns — which is why F1 and C1 exist at all. And it prices against its own rate configuration, not against 1,982 historical SDI jobs, the UDEF anchor, PMA_TBL, or the SDI workbook it has to mirror to the cell.

Worth knowing it exists, and worth being able to say precisely why we are not buying it.

→ **NO ACTION. AMMUNITION FOR THE QUESTION "WHY DID WE BUILD THIS?"**

Document AI has moved past pdfplumber

SECONDARY

Surya

LLMWhisperer

Camelot

The category now calls itself **Agentic Document Processing** — multimodal models plus layout awareness plus semantic reconstruction, rather than rule-based parsing.

The distinction that matters to us: tools that *dump* table data versus tools that produce **structured, labelled** output — knowing that one column holds part numbers, another quantities, another unit prices. That is precisely what `bom_table_extractor.py` is trying to do with rules.

The specific gap: pdfplumber and Camelot work well on digital PDFs with visible borders and *fail on scanned documents and borderless tables*. Layout-model tools like **Surya** and **LLMWhisperer** handle both. If any part of the BOM-table extraction is imperfect on customer-supplied or scanned packs, that is the shape of the fix — not a better regex.

→ **WHEN A BOM TABLE READS BADLY, CHECK FIRST WHETHER IT IS SCANNED OR BORDERLESS**

The architecture is already the recommended one

NO CHANGE NEEDED

The current best-practice advice for document extraction is a **hybrid**: rule-based parsers for structured, predictable fields, and models reserved for free-form content and edge cases where rigid rules fail.

That is the engine. Deterministic readers hold ranks 70-90, vision sits at rank 40 filling gaps, and the arbitration rule is fill-a-gap-always, replace-a-stronger-source-never.

Worth stating plainly because it is the sort of thing that gets questioned in a year when somebody suggests replacing the readers with a model. The industry consensus is that this would be the wrong move, and we already do the right one.

WHAT TO ACTUALLY DO

In order. Nothing here is more than a day's work.

#	ACTION	EFFORT	WHY NOW
1	Open SOLIDWORKS Labs, try LEO on a legacy PDF	An hour	F1 is on the roadmap as "POC pending". If LEO does it on a seat we already hold, that changes what gets built. Cheapest possible test of the most expensive roadmap item.
2	Point the Anthropic path at <code>claude-opus-5</code>	One line	Already wired, already keyed. Better instruction-following on the price fallback, and the effort toggle gives per-call cost control we do not have today.
3	Bump the xAI model string to Grok 4.6	One line	Same API shape. Do it <i>after</i> action 2 has been measured, not alongside — two changes in one run tells you nothing.
4	Read how Anchorpoint handles	An hour	P3.1 development starts 16 September. This is a storage- architecture decision

#	ACTION	EFFORT	WHY NOW
	SolidWorks binary churn		that is cheap now and expensive in January.
5	Price a 24 GB card; benchmark Qwen3.8-27B on 10575-02	A day	The only item here that costs money. It is also the only one that could remove per-call vision cost permanently and keep customer drawings on the network.
6	Price CADshift against keeping C1's DXF export in-house	20 minutes	A partner add-in already batch-exports laser-ready DXF with bend lines. It still needs a seat, so it is not a way round the licence — but it may mean not maintaining that half ourselves.
7	Note the Gemini introductory pricing end date	Minutes	\$0.75/\$3.75 reverts to \$1.50/\$7.50 on 31 December 2026. Any cost case built on the intro rate needs a January line.

What is deliberately not in this report

Nothing here proposes changing the engine's architecture, and nothing proposes replacing a deterministic reader with a model. The price chain works because steps 1 to 6 are reproducible and step 7 is flagged; every item above either sits at rank 40 where it cannot override a stronger source, or is a like-for-like model-string swap. **The interesting technology this month does not change that rule, and should not.**

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Compiled from web research on 26 August 2026. Vendor-sourced entries: solidworks.com (LEO, AURA, MARIE). Secondary sources include llm-stats.com, aireleasetracker.com, simonwillison.net, benchlm.ai, codersera.com,

yottalabs.ai, hawkridgesys.com, goengineer.com, anchorpoint.app, cadshift.com, digitalapplied.com, constructconnect.com and arxiv.org.

Figures marked Secondary are third-party reports and are not vendor-confirmed. Model pricing and availability move quickly; confirm against the vendor's own pricing page before committing spend or hardware. Introductory rates carry end dates.